25901814 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCZGY100, MG2DSCZIM100 - ENVIRONMENTAL BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. The South American low grasslands are known as …….. . [K] / [CO1]

   a). Pampas b). Steppes

   c). Prairies d). Savannas

2. …….. is the author of the famous book 'Man and Nature'. [K] / [CO1]

   a). Charles Darwin b). Rachel Carson

   c). George Perkins Marsh d). Ernst Haeckel

3. Select the primary ecological role of a predator. [U] / [CO2]

   a). Increasing prey populations b). Eliminating predators from ecosystems

   c). Controling prey populations and maintaining balance d). Providing shelter for other organisms

4. Name an effect of air pollution. [K] / [CO3]

a). Acid rain b). Oil Spills c). Hearing loss d). None of the above

5. Identify the primary function of the National Biodiversity Authority under the National Biodiversity Strategy and Action Plan. [U] / [CO3]

a). Formulating national policies b). Regulating biological resource access

c). Conducting agricultural research d). Constructing national parks

6. Expand WWF: [K] / [CO3]

a). World Wildlife Fund b). Wildlife World Fund

c). Wild World Fund for animals d). World Wide Fund

7. A rapid and intense flood that occurs within six hours of heavy rain: [K] / [CO4]

a). Coastal Flood b). Pluvial Flood

c). Urban Flood d). Flash Flood

8. The Environment Protection Act, 1986, empowers the government to create rules related to which of the following? [K] / [CO4]

a). Air and water pollution b). Hazardous waste management

c). Noise pollution d). All of the above

9. The primary challenge in implementing solid waste segregation: [U] / [CO4]

a). Lack of technology b). Public awareness and participation

c). High cost of bins d). Lack of available waste

10. The organization which plays a significant role in watershed development program in India: [K] / [CO4]

a). NASA b). IWDP

c). WTO d). FIFA

11. Bhopal Gas tragedy was caused by …….. . [K] / [CO4]

a). MIC b). Radioactive materials

c). Endosulfan d). Mercuric Iodide

12. Kyoto Protocol was aimed at …….. . [K] / [CO4]

a). Reduction of GHG emissions b). Protection of Ozone layer

c). Wetland conservation d). Disaster management

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. Discuss how human activities impact decomposers in an ecosystem. [U] / [CO1]

14. Distinguish between food chain and food web with examples. [U] / [CO1]

15. Differentiate between absolute natality and ecological natality. [U] / [CO2]

16. Describe exponential growth curve. [U] / [CO2]

17. Examine the role of women in the Chipko Movement. [U] / [CO3]

18. List 3 types of human activities allowed in wildlife sanctuaries. [K] / [CO3]

19. India is a mega-diversity nation. Explain. [U] / [CO3]

20. Interpret what a high Simpson's index value reveals about a community. [U] / [CO3]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21. Discuss the roles of bacteria in maintaining the nitrogen cycle. [U] / [CO1]

22. Compare and differentiate keystone species and umbrella species. [U] / [CO2]

23. Assess how the UN's Sustainable Development Goal 15 of the 2030 Agenda addresses biodiversity loss and ecosystem restoration. [An] / [CO3]

24. Examine how biodiversity conservation benefits human society and provide examples. [A] / [CO3]

25. Explain how the Forest Conservation Act, 1980, protects forest areas from unauthorized use or destruction. [U] / [CO4]

26. Explain the role of carbon trading in achieving national or global carbon reduction targets. [U] / [CO4]

[5x4 = 20]